

Lecture 2: Asymptotics

Jeremy Majerovitz (University of Notre Dame)
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Simulation

For each simulation:

- Uniform random variable: $X_i \sim [0, 1]$
- Repeat for $X_1, \dots, X_n$
- X_i is drawn i.i.d.
- Compute $\bar{X}_n = \frac{1}{n} \cdot \sum_{i=1}^n X_i$

Repeat simulation 10,000 times

Plot distribution of $\bar{X}_n$ across simulations

Distribution of Sample Mean, $n = 10$

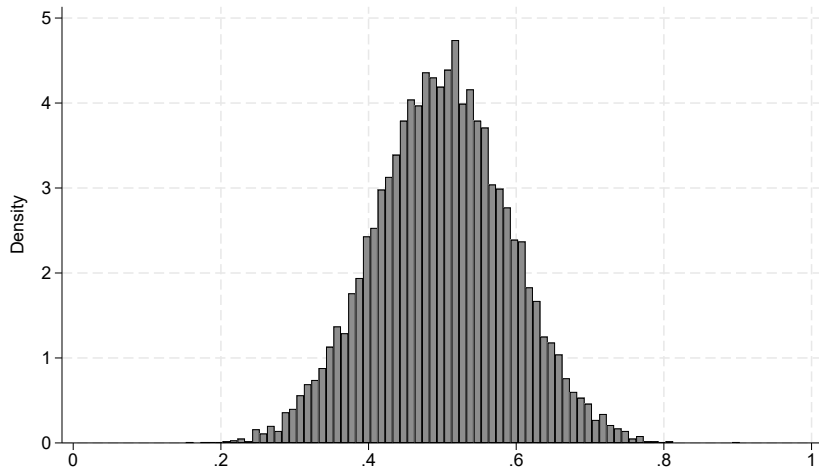


Figure 1: $\bar{X}_n, n = 10$

Distribution of Sample Mean, $n = 100$

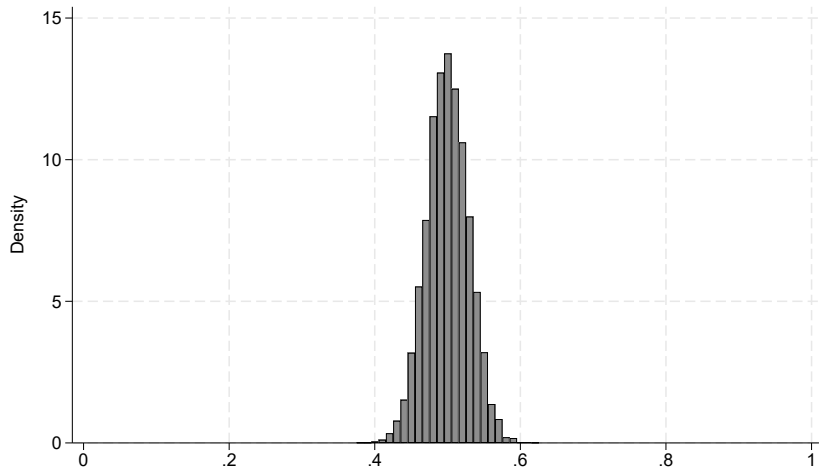


Figure 2: $\bar{X}_n$, $n = 100$

Distribution of Sample Mean, $n = 1,000$

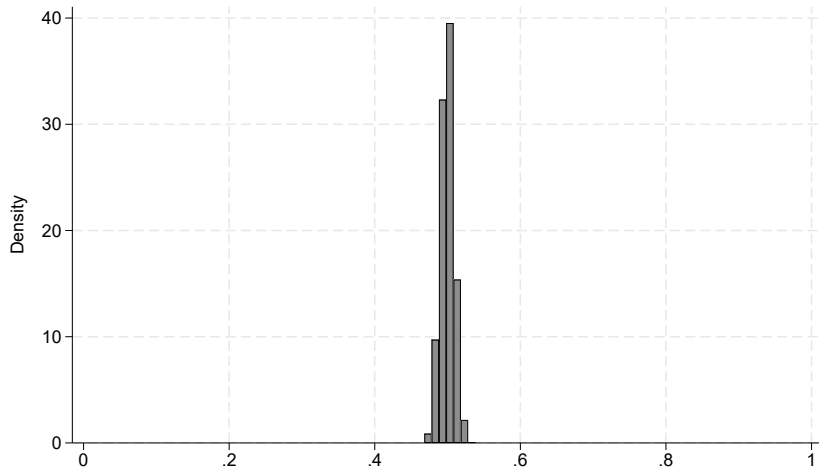


Figure 3: $\bar{X}_n$, $n = 1,000$

Distribution of Sample Mean, $n = 10,000$

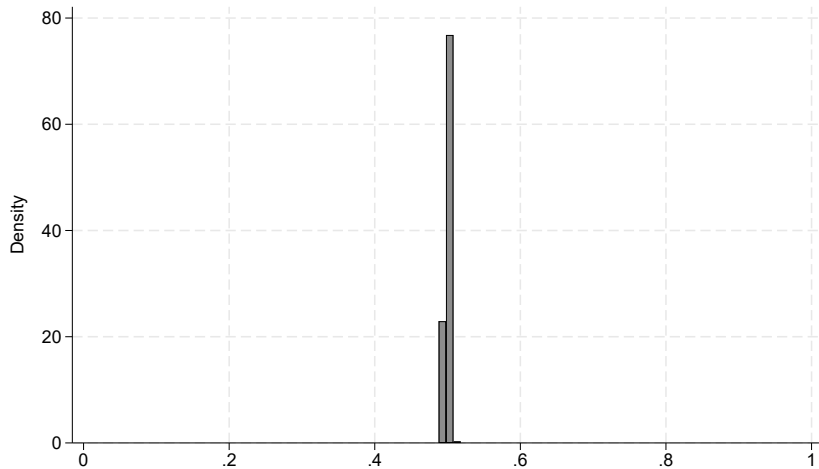


Figure 4: $\bar{X}_n$, $n = 10,000$

Scaled Mean

Next standardize the mean

$$\frac{\bar{X}_n - \mathbb{E}[\bar{X}_n]}{\sqrt{\text{Var}(\bar{X}_n)}} = \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$$

For $X_i \sim [0, 1]$, we have $\mu = 0.5$ and $\sigma = \sqrt{1/12}$

Repeat simulation 10,000 times

Plot distribution of $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$ across simulations

- Compare to pdf of the standard normal distribution

Distribution of Standardized Sample Mean, $n = 2$

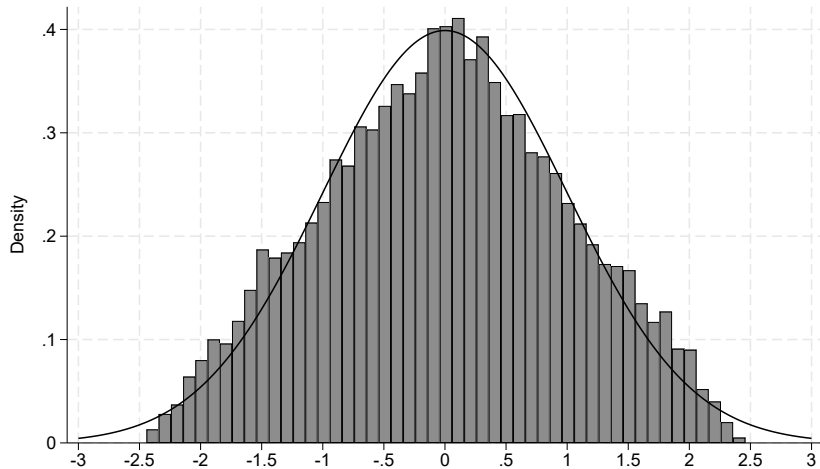


Figure 5: $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, $n = 2$

Distribution of Standardized Sample Mean, $n = 5$

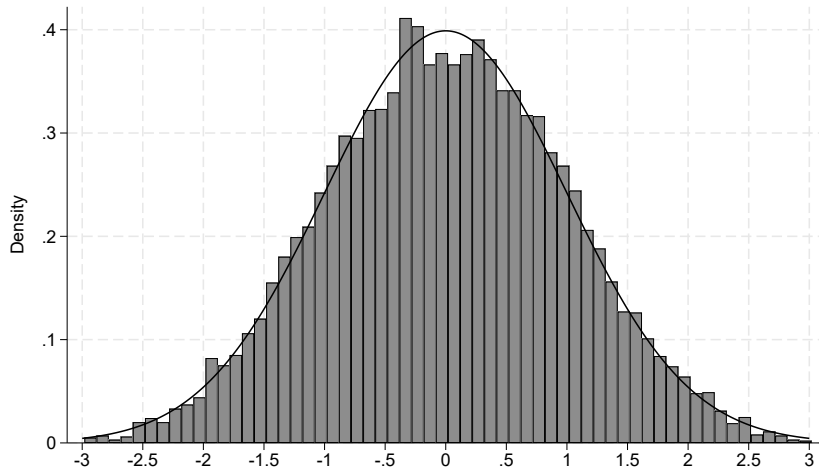


Figure 6: $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, $n = 5$

Distribution of Standardized Sample Mean, $n = 10$

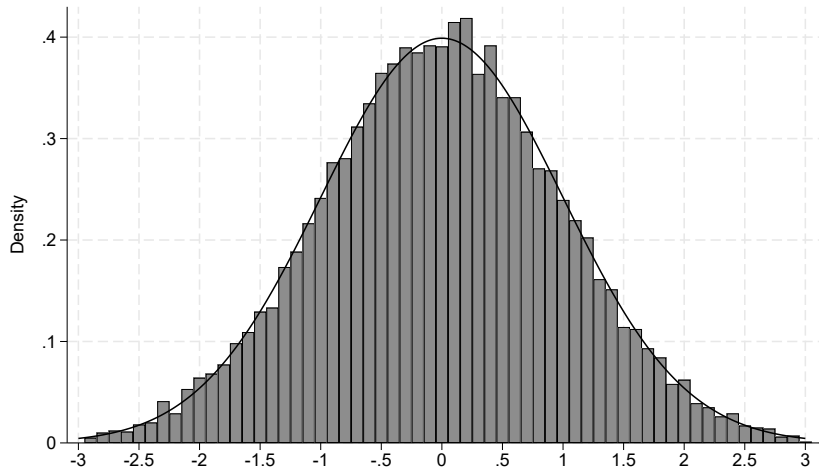


Figure 7: $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, $n = 10$

Distribution of Standardized Sample Mean, $n = 50$

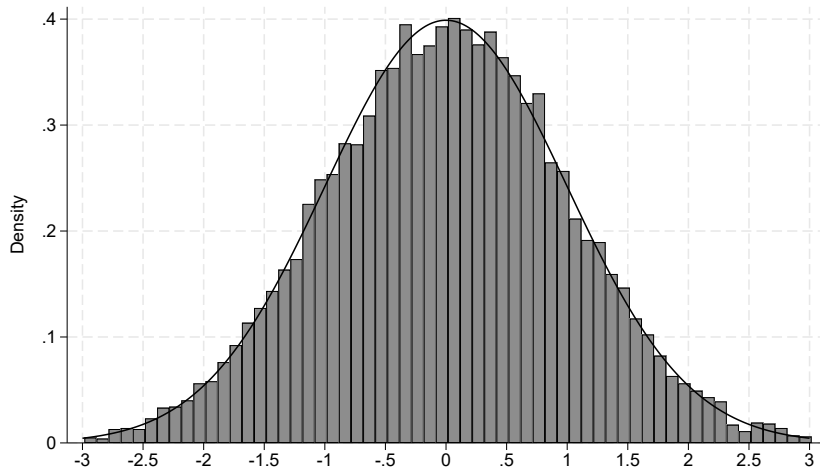


Figure 8: $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$, $n = 50$